

Domain-Adaptive Selective State-Space Learning for Cross-Subject Motor-Imagery EEG Decoding

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Abstract

Cross-subject motor-imagery electroencephalography (MI-EEG) decoding is hindered by low signal-to-noise ratios, scarce labeled data, and strong distribution shifts between subjects and sessions. This study proposes a domain-adaptive framework that couples a multi-scale convolutional front-end with bidirectional selective state-space blocks to capture both local and long-range temporal patterns, and reduces subject shift through hierarchical alignment, including a learned source–target normalization layer and label-free test-time adaptation. The framework was evaluated under a chronological leave-one-subject-out protocol in which target labels were never used for training, adaptation, or model selection. On BCI Competition IV-2a, it reached a mean accuracy of 75.81%, significantly higher than its source-only variant (64.35%; $p = 0.0039$) and its variant without test-time adaptation (72.15%; $p = 0.0078$). It also achieved 83.41% on BCI Competition IV-2b and 84.33% on Zhou2016, where it outperformed a matched TCFormer baseline on all four target subjects. Ablation studies showed that the contribution of each adaptation component depends on the dataset. These results indicate that combining selective state-space modeling with learned, hierarchical alignment improves cross-subject MI decoding without target labels.

Keywords: motor imagery, electroencephalography, brain–computer interface, selective state-space model, Mamba, domain adaptation, test-time adaptation

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1. Introduction

Motor-imagery brain-computer interfaces (MI-BCIs) decode movement intentions from non-invasive electroencephalography (EEG) for rehabilitation, assistive communication, and human-machine control. Their principal signatures, event-related desynchronization and synchronization, are weak, non-stationary, and distributed across time and electrodes [1], and they vary substantially between subjects and recording sessions. A model trained on existing users therefore often degrades on a new user, while collecting labeled calibration trials for every user is time-consuming. Cross-subject decoding must learn discriminative spatial-temporal structure from limited EEG and control inter-subject shift without target labels.

Existing work addresses this problem only partly. Convolutional and attention-based decoders learn strong spatial-temporal features but are trained on source subjects alone, whereas domain-adaptation methods reduce subject shift but are usually attached to generic backbones and evaluated under protocols in which the use of target-session data is not always explicit (Section 2).

We therefore propose a cross-subject MI-EEG framework that couples an EEG-specific selective state-space encoder with hierarchical, label-free adaptation, and evaluate it under a chronological leave-one-subject-out (LOSO) protocol in which target labels remain hidden throughout preprocessing, training, checkpoint selection, and adaptation. The contributions are:

- an end-to-end decoder that combines multi-scale convolution, alternating forward/backward selective state-space blocks, and local-global TCN fusion;
- a hierarchical adaptation strategy that combines Riemannian alignment, an AutoDIAL normalization layer, adversarial and MK-MMD alignment, and constrained test-time adaptation within a chronological, leakage-controlled LOSO protocol; and
- an evaluation on three public MI-EEG datasets with subject-wise accuracy, Cohen’s κ , statistical tests, efficiency analysis, and component ablations.

2. Related Work

2.1. Convolutional and attention-based MI decoders

Classical pipelines rely on common spatial pattern (CSP) and filter-bank CSP (FBCSP), which optimize class-specific variance in manually designed stages [2, 3]; their subject-specific covariance structure transfers poorly across users. Deep and shallow ConvNets and EEGNet learn temporal and electrode filters end to end [4, 5], but convolutions connect distant time points only through depth, large kernels, or dilation. Hybrid decoders therefore add global sequence modeling: EEG Conformer applies self-attention after convolutional tokenization, ATCNet combines attention with a dilated temporal convolutional network (TCN), and TCFormer integrates a multi-kernel front-end, grouped-query Transformer layers, and a grouped TCN head [6, 7, 8]. Self-attention, however, has quadratic sequence complexity [9] and can be data-hungry in the small-sample regimes typical of BCI studies.

2.2. Selective state-space models

State-space models offer linear-complexity sequence modeling. S4 introduced structured long-range state spaces [10], and Mamba made the transition quantities input dependent so that relevant information is retained selectively [11]. Recent MI decoders combine convolution with Mamba [12, 13], but they primarily study the architecture rather than its leakage-controlled integration with target adaptation, and a purely forward scan imposes unnecessary directionality when complete offline trials are available.

2.3. Cross-subject domain adaptation

Domain generalization learns subject-invariant representations from source subjects only [14, 15, 16], whereas unsupervised domain adaptation (UDA) also exploits unlabeled target EEG. UDA methods reduce covariance mismatch through Riemannian or Euclidean alignment [17, 18], minimize feature discrepancy with MK-MMD [19], or promote domain confusion through gradient reversal [20, 21]. HADANet combines adversarial and MK-MMD alignment for MI-EEG [22], and prototype-based methods additionally exploit class structure [23]. Alignment can also act inside normalization layers: AutoDIAL standardizes target features with a learned mixture of source and target statistics [24]. Marginal alignment nevertheless remains vulnerable to negative transfer, because it may mix classes and, when applied only to pooled trial features, can overlook temporal mismatch.

2.4. Test-time adaptation

Residual shift can also be reduced at inference. Information maximization encourages confident but class-diverse predictions without labels [25], and Tent restricts adaptation to normalization parameters [26]. Such constraints suit small EEG target sets, where unrestricted fine-tuning can overfit or collapse to one class, but they access evaluation-session signals and therefore require explicit disclosure and strict separation from labels.

2.5. Research gap

Three gaps follow. First, selective state-space MI decoders have not been combined with target adaptation, and their forward-only scans ignore the off-line availability of complete trials. Second, EEG adaptation methods align features with a fixed strength and mostly at the pooled-trial level; they neither learn how far target normalization should depart from source statistics nor align temporal representations. Third, the unlabeled target data used for adaptation are not always separated from, or disclosed relative to, the evaluation data. The proposed framework addresses these gaps with a bidirectional selective state-space encoder (Section 3.4), a hierarchical adaptation scheme that includes AutoDIAL normalization and temporal MK-MMD (Sections 3.6–3.8), and a chronological LOSO protocol that discloses every use of unlabeled target data (Section 4.2).

3. Proposed Method

Figure 1 summarizes one LOSO fold. A shared encoder processes labeled source trials and unlabeled target trials during Stage 1 UDA; an AutoDIAL layer then normalizes the pooled trial features before the residual aligner, classifier, and alignment losses. After source-validation checkpoint selection, Stage 2 adapts only normalization parameters with unlabeled evaluation-session EEG and re-estimates the AutoDIAL target statistics; labels are revealed only for Stage 3 scoring.

3.1. Problem formulation

Let $\mathcal{S} = \{(X_i^s, y_i^s)\}_{i=1}^{N_s}$ be labeled trials from the source subjects, $\mathcal{T}_a = \{X_j^a\}_{j=1}^{N_a}$ the earlier unlabeled target sessions used for UDA, and $\mathcal{T}_e = \{X_j^e\}_{j=1}^{N_e}$ the later target sessions used for evaluation, where $X \in \mathbb{R}^{C \times T}$. Training uses $\mathcal{S}$ and $\mathcal{T}_a$; IM-TTA may access only the signal values of $\mathcal{T}_e$, never their labels.

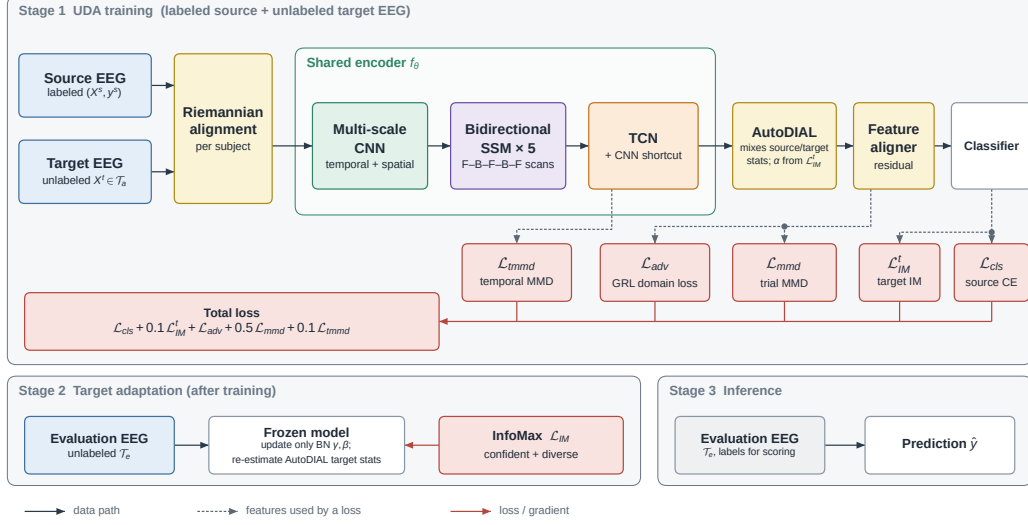


Figure 1: Framework for one LOSO fold: joint source classification and target adaptation with AutoDIAL feature normalization (Stage 1), label-free IM-TTA followed by AutoDIAL target-statistic re-estimation on the evaluation sessions $\mathcal{T}_e$ (Stage 2), and scoring with previously withheld labels (Stage 3).

3.2. Leakage-safe signal alignment

A separate Riemannian whitening operator is optionally fitted for each subject. OAS trial covariances are summarized by their affine-invariant mean $\bar{R}$, and trials are transformed as

$$\tilde{X} = \bar{R}^{-1/2} X. \quad (1)$$

A source transform is fitted on that subject’s training sessions and applied to its validation sessions; the target transform is fitted on $\mathcal{T}_a$ and frozen before application to $\mathcal{T}_e$ [17, 18]. Evaluation-session data therefore update neither the Riemannian transforms nor the input z-score statistics; the only statistics estimated on $\mathcal{T}_e$ are the AutoDIAL target moments in Stage 2 (Section 3.8).

3.3. Multi-scale convolutional encoder

As detailed in Fig. 2, three temporal convolutions with kernels of 20, 32, and 64 samples produce 32 maps each [8]. Their concatenated output is processed by depthwise convolution across all electrodes with depth multiplier two, average pooling, and dropout. A grouped pointwise projection reduces

the 192 spatial maps to 48 channels (16 per temporal scale), followed by grouped temporal convolution and residual channel-group attention. The latter assigns one learned coefficient to each scale. A second pooling layer yields $H_c \in \mathbb{R}^{B \times 48 \times L}$.

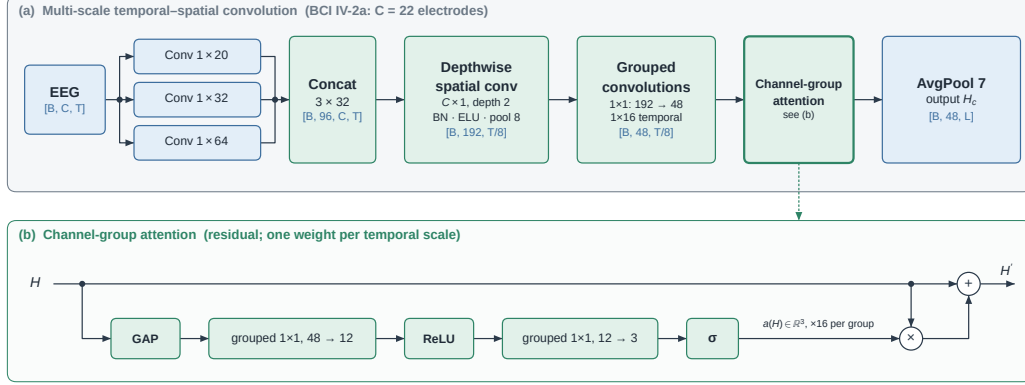


Figure 2: Multi-scale temporal-spatial front-end. (a) Three temporal branches, depthwise electrode filtering, grouped refinement, and channel-group attention produce H_c . (b) Channel-group attention: grouped 1×1 squeeze-excitation yields one sigmoid weight per temporal scale, which rescales that scale’s 16 channels and is added residually.

3.4. Alternating selective state-space encoder

After pointwise channel mixing, five selective state-space blocks scan in the order forward-backward-forward-backward-forward. A backward block reverses the temporal axis before and after its recurrence, providing bidirectional offline context without duplicating every block (Fig. 3). Within each block, layer normalization, gated projections, and a causal depthwise convolution generate u_t . A diagonal selective recurrence then computes [10, 11]

$$A = -\exp(A_{\log}), \quad (2)$$

$$h_t = \exp(\Delta_t A) \odot h_{t-1} + \Delta_t B_t \odot u_t, \quad (3)$$

$$z_t = C_t^\top h_t + d_{\text{skip}} \odot u_t. \quad (4)$$

Here Δ_t , B_t , and C_t depend on the input. The implementation uses state dimension eight, dropout 0.4, and depth-dependent stochastic depth. It is a lightweight PyTorch selective SSM rather than the official hardware-aware Mamba kernel, but retains linear scaling in sequence length.

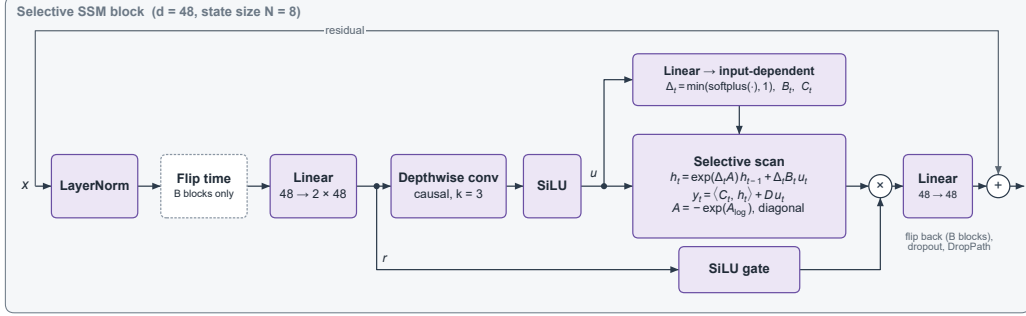


Figure 3: Selective SSM block. Backward blocks reverse and restore the temporal axis around the mixer; forward blocks omit these operations.

3.5. Local-global fusion and classification

The state-space output is reduced from 48 to 16 channels and concatenated with the 48-channel shortcut H_c . Two residual grouped TCN blocks with kernel length four and dilations one and two refine the resulting 64-channel sequence. The mean of its last state and temporal average forms the trial representation. A norm-constrained grouped 1×1 classifier produces four group-wise logits, which are combined by learned softmax weights.

3.6. Automatic domain alignment layer

Let $q \in \mathbb{R}^{64}$ denote a pooled trial representation, formed by the 16-channel groups of the three convolutional scales and the state-space branch. Following AutoDIAL [24], an alignment layer standardizes source features with source statistics (μ_s, σ_s^2) , whereas target features are standardized with the moments of the source-target mixture weighted by α :

$$\mu_\alpha = \alpha\mu_t + (1 - \alpha)\mu_s, \quad (5)$$

$$\sigma_\alpha^2 = \alpha\sigma_t^2 + (1 - \alpha)\sigma_s^2 + \alpha(1 - \alpha)(\mu_t - \mu_s)^2, \quad (6)$$

$$\hat{q}^s = \gamma \odot \frac{q^s - \mu_s}{\sqrt{\sigma_s^2 + \epsilon}} + \beta, \quad \hat{q}^t = \gamma \odot \frac{q^t - \mu_\alpha}{\sqrt{\sigma_\alpha^2 + \epsilon}} + \beta. \quad (7)$$

The affine parameters γ and β are shared by both domains. A separate $\alpha = \text{sigmoid}(a) \in (0, 1)$, initialized to 0.5, is learned for each of the four feature groups: $\alpha \rightarrow 0$ normalizes the target with source statistics, whereas $\alpha \rightarrow 1$ recovers fully domain-specific normalization. During training, mini-batch moments of each domain are used and running estimates are updated

with momentum 0.1; at inference, the running source and target moments are mixed with the learned α . Source validation trials are always normalized with source statistics.

3.7. Hierarchical unsupervised domain adaptation

A residual $64 \rightarrow 128 \rightarrow 64$ feature aligner follows the AutoDIAL layer and precedes classification and domain alignment. Source cross-entropy $\mathcal{L}_{\text{cls}}$ maintains discrimination. A gradient-reversal discriminator provides domain loss $\mathcal{L}_{\text{adv}}$ [20], while five-kernel MK-MMD aligns normalized source and target representations [19]. It is applied both to pooled trial features ($\mathcal{L}_{\text{mmd}}$) and to time-averaged TCN sequences ($\mathcal{L}_{\text{tmmd}}$). In addition, the information-maximization objective $\mathcal{L}_{\text{IM}}$ of Eq. (9) is evaluated on the predictions for the unlabeled target mini-batch ($\mathcal{L}_{\text{IM}}^t$), giving

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + 0.1\mathcal{L}_{\text{IM}}^t + \mathcal{L}_{\text{adv}} + 0.5\mathcal{L}_{\text{mmd}} + 0.1\mathcal{L}_{\text{tmmd}}. \quad (8)$$

The alignment losses $\mathcal{L}_{\text{adv}}$, $\mathcal{L}_{\text{mmd}}$, and $\mathcal{L}_{\text{tmmd}}$ see the AutoDIAL output with α held fixed, because they could otherwise be reduced trivially by driving α toward one. Since source features do not depend on α , the mixing weights are learned only through $\mathcal{L}_{\text{IM}}^t$, that is, toward the normalization that yields confident yet class-diverse target predictions. Only $\mathcal{L}_{\text{cls}}$ uses labels, all of which come from source subjects; target trials in this stage come exclusively from $\mathcal{T}_a$. The discriminator coefficient increases smoothly from zero to one with training progress.

3.8. Constrained information-maximization adaptation

After checkpoint selection, all parameters except BatchNorm affine weights and biases are frozen. Five Adam passes over unlabeled $\mathcal{T}_e$ at learning rate 10^{-4} minimize

$$\mathcal{L}_{\text{IM}} = \frac{1}{N_e} \sum_j H(p_j) - H\left(\frac{1}{N_e} \sum_j p_j\right), \quad (9)$$

where the first term encourages confident predictions and the second preserves class diversity [25, 26]. Because IM-TTA changes the backbone features, the AutoDIAL target moments (μ_t, σ_t^2) are then re-estimated in one pass over the same unlabeled $\mathcal{T}_e$ trials, while the learned α and the source moments remain fixed. No evaluation label or metric is used for adaptation, re-estimation, or checkpoint selection.

4. Experimental Setup

4.1. Datasets

Three public MI-EEG datasets with different numbers of electrodes, classes, subjects, and sessions are used (Table 1). All are accessed through MOABB/Braindecode, which provides a common representation of subjects, sessions, runs, and event windows [27].

Table 1: Summary of the MI-EEG datasets (EOG channels excluded).

Dataset	Subjects	Sex (M/F)	Channels	Sessions	Classes
BCI IV-2a	9	n.r.	22	2	left hand, right hand, feet, tongue
BCI IV-2b	9	n.r. ^a	3	5	left hand, right hand
Zhou2016	4	1/3	14	3	left hand, right hand, feet

n.r.: not reported in the dataset description. ^aThe nine participants were drawn from a study of ten (6/4, 24.7 ± 3.3 years) [28]; the split of the nine released participants is not reported.

4.1.1. BCI Competition IV-2a

The dataset contains nine participants recorded in two sessions on different days; each session has 288 trials distributed equally over imagined left-hand, right-hand, both-feet, and tongue movements. EEG was recorded from 22 electrodes (three additional EOG channels are not used) at 250 Hz with hardware band-pass filtering at 0.5–100 Hz and a 50-Hz notch [29]. The two sessions are denoted session 1 and session 2. Participant sex and age are not reported.

4.1.2. BCI Competition IV-2b

The dataset contains two-class left- versus right-hand MI from nine right-handed participants with normal or corrected-to-normal vision, drawn from a study of ten participants (six male, four female; age 24.7 ± 3.3 years) [28]; the sex of the nine released participants is not reported individually. Three bipolar derivations around C3, Cz, and C4 (three EOG channels are not used) were sampled at 250 Hz. Sessions 1–3 form the original calibration portion and sessions 4–5 the evaluation portion [29], a chronological separation that is kept here. The dataset tests whether the method remains useful with only three electrodes.

4.1.3. Zhou2016

The dataset contains four healthy participants (one male and three female, aged 22–28 years; sex reported only at group level), each recorded in three sessions separated by days to months [30]. Each session has two runs of 75 trials (25 per class: left hand, right hand, and feet), recorded from 14 electrodes at 250 Hz. Each trial contains a 5-s cue period, during which the imagery is performed, followed by a 4-s rest.

4.2. Chronological LOSO protocol with test-time adaptation

One subject is held out in each LOSO fold. All remaining subjects are sources; the held-out participant is the target. Table 2 specifies the chronological session split. Earlier target sessions are used without labels during Stage 1 UDA. The later evaluation sessions remain unavailable during model training, preprocessing-statistic fitting, and checkpoint selection; at inference time, their EEG signals are passed without labels to IM-TTA and AutoDIAL target-statistic re-estimation, after which the adapted model is scored using the withheld labels. This is the primary protocol for headline comparisons.

Table 2: Chronological session allocation in each LOSO fold. Evaluation-session EEG is available without labels only to IM-TTA and AutoDIAL target-statistic re-estimation at inference time; its labels are used afterward solely for scoring.

Dataset	Source subjects (labeled)		Target subject	
	Training	Validation	Stage 1 UDA (unlabeled)	IM-TTA + evaluation (labels withheld)
BCI IV-2a	session 1	session 2	session 1	session 2
BCI IV-2b	sessions 1–3	sessions 4–5	sessions 1–3	sessions 4–5
Zhou2016	sessions 1–2	session 3	sessions 1–2	session 3

This split yields nine folds for BCI IV-2a and BCI IV-2b and four for Zhou2016, with identical split rules and hyperparameters in every fold. The checkpoint with the highest source-validation accuracy is selected before IM-TTA, and neither evaluation-session labels nor evaluation metrics participate in adaptation or checkpoint selection. Stage 1 adaptation (AutoDIAL, target information maximization, and the adversarial and MK-MMD losses) uses only the earlier target sessions, whereas Stage 2 IM-TTA and AutoDIAL re-estimation use the later evaluation sessions immediately before scoring. For

Zhou2016, an all-session target score is additionally reported as an auxiliary diagnostic only.

4.3. Preprocessing and augmentation

Only EEG channels are retained. Signals are resampled to 250 Hz and converted from volts to microvolts. The BCI Competition datasets receive no additional digital band-pass filtering: BCI IV-2a uses the complete MOABB event window, while BCI IV-2b removes 0.5 s from its end. For Zhou2016, all 14 EEG channels are band-pass filtered at 8–30 Hz and the interval from 0.5 to 5.0 s after cue onset is retained (4.5 s, or 1125 samples). This setting suppresses the initial cue transient and focuses both feature extraction and covariance estimation on the motor-imagery μ/β range.

Riemannian whitening, when active, follows the session-isolated fitting rule in Section 3.2. Z-score statistics are fitted using the source-training partition and then frozen. Following TCFormer [8], the segmentation and reconstruction (S&R) augmentation introduced by Lotte [31] is applied to labeled source mini-batches of BCI IV-2a and IV-2b: each trial is divided into eight consecutive, non-overlapping segments, and synthetic trials are formed by concatenating, for each segment position, the segment of a randomly chosen trial of the same class, which doubles the mini-batch. It neither reads target labels nor mixes source and target trials. Zhou2016 is run without this augmentation to preserve its intended dataset-specific setting.

4.4. Implementation details

All experiments are executed in Kaggle notebooks using one NVIDIA Tesla T4 GPU with 16 GB memory. BCI IV-2a and BCI IV-2b use a mini-batch size of 48 and a maximum of 125 epochs. Zhou2016 uses a mini-batch size of 32 and 100 epochs. The BCI Competition runs use Adam with an initial learning rate of 9×10^{-4} , $\beta_1 = 0.5$, and weight decay 10^{-3} ; the Zhou2016 setting uses an initial learning rate of 10^{-3} and weight decay 10^{-4} . A linear warm-up over the first three LOSO epochs is followed by cosine decay. The random seed (0), data split, and initialization procedure are fixed and shared by all methods and ablation variants, so that every reported comparison differs only in the evaluated component.

4.5. Evaluation measures and reporting

Accuracy and Cohen’s κ ($\times 100$) are reported per target subject and as mean $\pm$ standard deviation across LOSO folds. Paired accuracy differences

use a two-sided Wilcoxon signed-rank test with exact sign-permutation p -values. Efficiency is summarized by trainable parameters, mean Stage 1 training time per fold on a Tesla T4, and batch-one CPU response time averaged over 500 passes after 100 warm-up passes; IM-TTA, preprocessing, and data transfer are excluded from these timings.

4.6. Baseline comparison policy

All methods follow the same LOSO folds, chronological split, and target-data access in Table 2; results obtained under incompatible protocols are not copied. EEG Conformer [6], ATCNet [7], TCFormer [8], DAN [19], and CDAN [21] are the selected baselines. The proposed method is reported in three variants:

- **Ours-S**: source-only training with $\mathcal{L}_{\text{cls}}$;
- **Ours-U**: Stage 1 UDA, including AutoDIAL, without IM-TTA; the AutoDIAL target statistics are still re-estimated on unlabeled $\mathcal{T}_e$; and
- **Ours**: Ours-U followed by label-free IM-TTA on the evaluation-session EEG.

Thus, Ours-U–Ours-S isolates Stage 1 adaptation, while Ours–Ours-U isolates IM-TTA.

5. Results

Tables 3–5 report subject-wise chronological LOSO results. Standard deviations describe variation across target subjects, not repeated seeds.

5.1. BCI Competition IV-2a

Table 3 reports the four-class chronological-session LOSO results for the nine BCI IV-2a target subjects. Ours applies label-free IM-TTA to each target subject’s session 2 EEG immediately before evaluation.

Ours achieved $75.81 \pm 10.63\%$ accuracy and 67.70 ± 14.20 Cohen’s κ , outperforming Ours-S ($64.35 \pm 12.77\%$) and Ours-U ($72.15 \pm 13.58\%$). Both paired accuracy differences were significant ($p = 0.0039$ and $p = 0.0078$, respectively); IM-TTA improved eight of the nine target folds. As an auxiliary diagnostic only, Ours-U obtained $73.32 \pm 11.94\%$ accuracy and 64.40 ± 15.90 Cohen’s κ when target sessions 1 and 2 were pooled.

Table 3: Per-subject LOSO accuracy (%) and Cohen’s κ multiplied by 100 on BCI IV-2a (four classes). Ours-S: source-only; Ours-U: UDA without IM adaptation; Ours: UDA with IM adaptation (Section 4.6). p : Wilcoxon signed-rank test of accuracy against Ours.

Subj.		Source-only				UDA			
		Conformer	ATCNet	TCFormer	Ours-S	DAN	CDAN	Ours-U	Ours
		[6]	[7]	[8]		[19]	[21]		
S1	Acc.	—	—	—	68.06	—	—	79.17	78.82
	κ	—	—	—	57.41	—	—	72.22	71.76
S2	Acc.	—	—	—	51.04	—	—	47.57	57.64
	κ	—	—	—	34.72	—	—	30.09	43.52
S3	Acc.	—	—	—	86.46	—	—	92.71	93.75
	κ	—	—	—	81.94	—	—	90.28	91.67
S4	Acc.	—	—	—	61.11	—	—	73.61	74.31
	κ	—	—	—	48.15	—	—	64.81	65.74
S5	Acc.	—	—	—	64.24	—	—	57.64	65.62
	κ	—	—	—	52.31	—	—	43.52	54.17
S6	Acc.	—	—	—	39.58	—	—	59.72	66.67
	κ	—	—	—	19.44	—	—	46.30	55.56
S7	Acc.	—	—	—	72.92	—	—	82.29	85.42
	κ	—	—	—	63.89	—	—	76.39	80.56
S8	Acc.	—	—	—	73.61	—	—	82.64	84.38
	κ	—	—	—	64.81	—	—	76.85	79.17
S9	Acc.	—	—	—	62.15	—	—	73.96	75.69
	κ	—	—	—	49.54	—	—	65.28	67.59
Mean	Acc.	—	—	—	64.35	—	—	72.15	75.81
	κ	—	—	—	52.50	—	—	62.90	67.70
SD	Acc.	—	—	—	12.77	—	—	13.58	10.63
	κ	—	—	—	17.00	—	—	18.10	14.20
p		—	—	—	0.0039	—	—	0.0078	—

5.2. BCI Competition IV-2b

Table 4 reports the completed binary left- versus right-hand results for the nine BCI IV-2b target subjects, whose recordings contain only three bipolar EEG derivations. Results use sessions 4–5 for the primary chronological evaluation; the standard deviation measures between-subject dispersion in this single LOSO run.

Ours achieved $83.41 \pm 10.35\%$ accuracy and 66.80 ± 20.70 Cohen’s κ , compared with $80.54 \pm 10.07\%$ for Ours-S and $83.97 \pm 10.71\%$ for Ours-U. Ours-U was 0.56 percentage points higher than Ours on average, but the paired difference was not significant ($p = 0.4063$); Ours likewise did not differ significantly from Ours-S ($p = 0.2500$). IM-TTA improved three of the nine target folds.

5.3. Zhou2016

Table 5 uses source sessions 1–2 for training, source session 3 for validation, target sessions 1–2 for UDA, and target session 3 for primary scoring. With four folds, the smallest attainable two-sided Wilcoxon p -value is 0.125.

Ours reached $84.33 \pm 0.75\%$ accuracy and 76.50 ± 1.12 Cohen’s κ , the highest mean and the lowest variability across targets. It exceeded the matched TCFormer baseline ($73.50 \pm 6.22\%$; $\kappa = 60.25 \pm 9.34$) and Ours-S ($75.17 \pm 3.51\%$; $\kappa = 62.75 \pm 5.26$) on all four targets ($p = 0.1250$, the smallest attainable value with four folds). Ours-U, which omits IM-TTA while retaining Stage 1 adaptation and AutoDIAL target-statistic re-estimation, reached $82.33 \pm 3.82\%$ ($\kappa = 73.50 \pm 5.70$; $p = 0.6250$). As an auxiliary diagnostic only, pooling all three target sessions gave $83.23 \pm 2.89\%$ ($\kappa = 74.85 \pm 4.34$) for Ours and $75.91 \pm 4.65\%$ ($\kappa = 63.90 \pm 7.00$) for TCFormer.

5.4. Ablation study

Table 6 follows the primary chronological protocol. “w/o DA” removes every Stage 1 domain-adaptation component (AutoDIAL, target information maximization, and the adversarial and both MMD losses) while retaining RA and IM-TTA; all variants otherwise use the same folds and checkpoint rule.

On BCI IV-2a, removing RA or DA reduced accuracy from 75.81% to 69.06% and 71.37%, respectively; the no-DA difference was significant ($p = 0.0078$). On BCI IV-2b, no-RA (83.61%), no-IM (83.97%; $p = 0.4063$), and no-DA (83.11%; $p = 0.4961$) were effectively comparable with the complete model (83.41%). On Zhou2016, the complete model (84.33%) was slightly above no-RA (84.00%), no-DA (82.50%), and no-IM (82.33%); with only four

Table 4: Per-subject LOSO accuracy (%) and Cohen’s κ multiplied by 100 on BCI IV-2b (two classes). Ours-S: source-only; Ours-U: UDA without IM adaptation; Ours: UDA with IM adaptation (Section 4.6). p : Wilcoxon signed-rank test of accuracy against Ours.

Subj.		Source-only				UDA			
		Conformer	ATCNet	TCFormer	Ours-S	DAN	CDAN	Ours-U	Ours
		[6]	[7]	[8]		[19]	[21]		
S1	Acc.	—	—	—	77.50	—	—	78.44	77.50
	κ	—	—	—	55.00	—	—	56.88	55.00
S2	Acc.	—	—	—	68.57	—	—	70.71	67.86
	κ	—	—	—	37.14	—	—	41.43	35.71
S3	Acc.	—	—	—	63.13	—	—	63.44	65.94
	κ	—	—	—	26.25	—	—	26.88	31.88
S4	Acc.	—	—	—	89.38	—	—	90.31	88.75
	κ	—	—	—	78.75	—	—	80.62	77.50
S5	Acc.	—	—	—	96.56	—	—	97.81	96.88
	κ	—	—	—	93.12	—	—	95.63	93.75
S6	Acc.	—	—	—	73.44	—	—	83.75	85.94
	κ	—	—	—	46.88	—	—	67.50	71.88
S7	Acc.	—	—	—	85.00	—	—	89.69	90.62
	κ	—	—	—	70.00	—	—	79.37	81.25
S8	Acc.	—	—	—	86.25	—	—	95.63	94.06
	κ	—	—	—	72.50	—	—	91.25	88.13
S9	Acc.	—	—	—	85.00	—	—	85.94	83.13
	κ	—	—	—	70.00	—	—	71.88	66.25
Mean	Acc.	—	—	—	80.54	—	—	83.97	83.41
	κ	—	—	—	61.10	—	—	67.90	66.80
SD	Acc.	—	—	—	10.07	—	—	10.71	10.35
	κ	—	—	—	20.10	—	—	21.40	20.70
p		—	—	—	0.2500	—	—	0.4063	—

Table 5: Per-subject LOSO accuracy (%) and Cohen’s κ multiplied by 100 on target session 3 of Zhou2016 (three classes) under the clean chronological protocol. Ours-S: source-only; Ours-U: UDA without IM adaptation; Ours: UDA with IM adaptation. Bold: best value in each row. p : Wilcoxon signed-rank test of accuracy against Ours.

Subj.		Source-only				UDA			
		Conformer	ATCNet	TCFormer	Ours-S	DAN	CDAN	Ours-U	Ours
		[6]	[7]	[8]		[19]	[21]		
S1	Acc.	–	–	68.00	72.00	–	–	79.33	84.00
	κ	–	–	52.00	58.00	–	–	69.00	76.00
S2	Acc.	–	–	78.67	78.67	–	–	84.67	83.33
	κ	–	–	68.00	68.00	–	–	77.00	75.00
S3	Acc.	–	–	66.67	71.33	–	–	87.33	84.67
	κ	–	–	50.00	57.00	–	–	81.00	77.00
S4	Acc.	–	–	80.67	78.67	–	–	78.00	85.33
	κ	–	–	71.00	68.00	–	–	67.00	78.00
Mean	Acc.	–	–	73.50	75.17	–	–	82.33	84.33
	κ	–	–	60.25	62.75	–	–	73.50	76.50
SD	Acc.	–	–	6.22	3.51	–	–	3.82	0.75
	κ	–	–	9.34	5.26	–	–	5.70	1.12
p		–	–	0.1250	0.1250	–	–	0.6250	–

Table 6: Ablation results under the primary chronological protocol: mean $\pm$ standard deviation of accuracy (%) and Cohen’s κ ($\times 100$) across target subjects. BCI IV-2a is scored on session 2, BCI IV-2b on sessions 4–5, and Zhou2016 on session 3. RA: Riemannian alignment; DA: Stage 1 domain adaptation; IM: IM-TTA.

Dataset	Metric	Complete	w/o RA	w/o DA	w/o IM
BCI IV-2a	Acc.	75.81 $\pm$ 10.63	69.06 $\pm$ 8.97	71.37 $\pm$ 11.60	72.15 $\pm$ 13.58
	κ	67.70 $\pm$ 14.20	58.70 $\pm$ 12.00	61.80 $\pm$ 15.50	62.90 $\pm$ 18.10
BCI IV-2b	Acc.	83.41 $\pm$ 10.35	83.61 $\pm$ 10.06	83.11 $\pm$ 11.47	83.97 $\pm$ 10.71
	κ	66.80 $\pm$ 20.70	67.20 $\pm$ 20.10	66.20 $\pm$ 22.90	67.90 $\pm$ 21.40
Zhou2016	Acc.	84.33 $\pm$ 0.75	84.00 $\pm$ 3.16	82.50 $\pm$ 1.85	82.33 $\pm$ 3.82
	κ	76.50 $\pm$ 1.12	76.00 $\pm$ 4.70	73.75 $\pm$ 2.77	73.50 $\pm$ 5.70

folds, none of these differences can be significant (the smallest attainable p is 0.125; $p = 0.2500$ for no-DA and 0.6250 for no-IM). Thus, the complete framework was best on BCI IV-2a and Zhou2016, whereas the variants were statistically indistinguishable on BCI IV-2b.

5.5. Computational efficiency

Table 7 reports trainable parameters, Stage 1 training time, and CPU response time. Ours-S is smaller because it omits the AutoDIAL layer, aligner, and domain discriminator; RA and IM-TTA do not change parameter count.

Because the token sequence after temporal pooling is short (20 tokens on Zhou2016), a step-by-step selective scan is dominated by per-operation dispatch overhead rather than arithmetic. All Zhou2016 runs of the proposed method therefore use a loop-hoisted scan: $\exp(\Delta_t A)$, $\Delta_t B_t \odot u_t$, and the C_t readout are computed for all time steps outside the recurrence, leaving one multiply-add per step, and the causal depthwise convolution is evaluated as shifted multiply-adds. This reorganization changes neither the model nor its parameters (output logits agree with the per-step scan to within 10^{-8}). Ours-S required 6.29 min per fold and 13.36 ms per CPU trial, compared with 11.42 min and 14.65 ms for Ours-U, 11.66 min and 18.88 ms for w/o RA, 6.46 min and 18.41 ms for w/o DA, and 11.70 min and 19.14 ms for the complete model.

Table 7: Computational cost: trainable parameters, mean Stage 1 training time per fold, and single-trial CPU response time. [†]Measured with the loop-hoisted selective scan (Section 5.5); the BCI IV-2a and IV-2b rows use the per-step scan, which computes the same function.

Dataset	Method	Params (K)	Train (min)	CPU (ms)
BCI IV-2a	Ours-S	98.4	47.01	25.08
	Ours-U	132.0	58.45	18.44
	Ours w/o RA	131.9	64.43	32.42
	Ours w/o DA	131.9	47.52	29.95
	Ours	131.9	63.58	28.98
BCI IV-2b	Ours-S	94.6	45.17	21.89
	Ours-U	128.2	48.08	13.35
	Ours w/o RA	128.1	58.09	23.05
	Ours w/o DA	128.1	45.75	23.34
	Ours	128.1	60.74	24.72
Zhou2016	TCFormer	125.6	5.89	15.97
	Ours-S [†]	96.8	6.29	13.36
	Ours-U [†]	130.4	11.42	14.65
	Ours w/o RA [†]	130.4	11.66	18.88
	Ours w/o DA [†]	130.3	6.46	18.41
	Ours[†]	130.4	11.70	19.14

6. Discussion

The results show that adaptation is dataset dependent. The complete framework was best on BCI IV-2a and Zhou2016, whereas on BCI IV-2b the no-IM variant was slightly better and all variants were statistically indistinguishable. On Zhou2016, the complete configuration also had the lowest variability across targets, but its margins over the variants without RA, IM-TTA, or Stage 1 DA were small and cannot be significant with four folds. Filtering to 8–30 Hz and excluding the first 0.5 s focus covariance estimation on MI-related activity rather than broadband drift and cue transients; repeated-seed experiments are still required to establish the stability of these differences.

Architecturally, multi-scale and depthwise convolutions preserve local spectral-spatial evidence, while alternating selective scans and the TCN model complementary long- and short-range dynamics. The alternating scans exploit complete offline trials and are non-causal; online use would require forward-only processing and device-specific latency validation.

The method is transductive because IM-TTA accesses unlabeled evaluation-session signals before scoring. Comparisons must therefore provide baselines with identical target access. Other limitations include the three-channel BCI IV-2b setting, the small Zhou2016 cohort, and the absence of repeated-seed and fully matched baseline results. Participant sex is reported only in aggregate or not at all for the public datasets, and the cohorts are too small for sex- or gender-stratified analysis; potential sex-related differences in decoding performance were therefore not assessed and limit the generalizability of the findings.

7. Conclusion

We presented a cross-subject MI-EEG framework combining multi-scale spatial-temporal convolution, alternating selective state-space blocks, local-global TCN fusion, hierarchical UDA, and constrained IM-TTA. Its chronological LOSO protocol excludes target labels from preprocessing, training, adaptation, and checkpoint selection. Ours obtained $75.81 \pm 10.63\%$ session-2 accuracy on BCI IV-2a and $83.41 \pm 10.35\%$ sessions 4–5 accuracy on BCI IV-2b, with corresponding Cohen’s κ scores of 67.70 ± 14.20 and 66.80 ± 20.70 on the $\times 100$ scale. On Zhou2016, Ours obtained $84.33 \pm 0.75\%$ target-session-3 accuracy and 76.50 ± 1.12 Cohen’s κ . Matched baselines and repeated-seed experiments remain necessary before drawing broader conclusions.

CRediT authorship contribution statement

[Author name]: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All EEG data analyzed in this study are publicly available. BCI Competition IV datasets 2a and 2b [29] and the Zhou2016 dataset [30] were accessed through the MOABB library [27]; no new data were collected.

References

- [1] G. Pfurtscheller, F. H. Lopes da Silva, Event-related EEG/MEG synchronization and desynchronization: basic principles, *Clinical Neurophysiology* 110 (11) (1999) 1842–1857. doi:10.1016/S1388-2457(99)00141-8.
- [2] H. Ramoser, J. Müller-Gerking, G. Pfurtscheller, Optimal spatial filtering of single trial EEG during imagined hand movement, *IEEE Transactions on Rehabilitation Engineering* 8 (4) (2000) 441–446. doi:10.1109/86.895946.
- [3] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, H. Zhang, Filter bank common spatial pattern algorithm on BCI competition IV datasets 2a and 2b, *Frontiers in Neuroscience* 6 (2012) 39. doi:10.3389/fnins.2012.00039.
- [4] R. T. Schirrmeister, J. T. Springenberg, L. D. J. Fiederer, M. Glasstetter, K. Eggenberger, M. Tangermann, F. Hutter, W. Burgard, T. Ball, Deep learning with convolutional neural networks for EEG decoding and visualization, *Human Brain Mapping* 38 (11) (2017) 5391–5420. doi:10.1002/hbm.23730.
- [5] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, B. J. Lance, EEGNet: A compact convolutional neural network for EEG-based brain–computer interfaces, *Journal of Neural Engineering* 15 (5) (2018) 056013. doi:10.1088/1741-2552/aace8c.

- [6] Y. Song, Q. Zheng, B. Liu, X. Gao, EEG Conformer: Convolutional transformer for EEG decoding and visualization, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 31 (2023) 710–719. doi:10.1109/TNSRE.2022.3230250.
- [7] H. Altaheri, G. Muhammad, M. Alsulaiman, Physics-informed attention temporal convolutional network for EEG-based motor imagery classification, *IEEE Transactions on Industrial Informatics* 19 (2) (2023) 2249–2258. doi:10.1109/TII.2022.3197419.
- [8] H. Altaheri, F. Karray, A.-H. Karimi, Temporal convolutional transformer for EEG based motor imagery decoding, *Scientific Reports* 15 (1) (2025) 32959. doi:10.1038/s41598-025-16219-7.
- [9] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, in: *Advances in Neural Information Processing Systems*, Vol. 30, 2017, pp. 5998–6008.
- [10] A. Gu, K. Goel, C. Ré, Efficiently modeling long sequences with structured state spaces, in: *International Conference on Learning Representations*, 2022.
URL <https://openreview.net/forum?id=uYLFoz1v1AC>
- [11] A. Gu, T. Dao, Mamba: Linear-time sequence modeling with selective state spaces, in: *First Conference on Language Modeling*, 2024.
URL <https://openreview.net/forum?id=tEYskw1VY2>
- [12] M. Guo, X. Han, H. Liu, J. Zhu, J. Zhang, Y. Bai, G. Ni, MI-Mamba: A hybrid motor imagery electroencephalograph classification model with Mamba’s global scanning, *Annals of the New York Academy of Sciences* 1544 (1) (2025) 242–253. doi:10.1111/nyas.15288.
- [13] X. Yang, Z. Jia, Spatial-temporal Mamba network for EEG-based motor imagery classification, in: *Advanced Data Mining and Applications, Lecture Notes in Computer Science*, Springer Nature Singapore, 2025, pp. 418–432. doi:10.1007/978-981-96-0821-8_28.
- [14] S. Liu, L. An, C. Zhang, Z. Jia, A spatial-temporal transformer based on domain generalization for motor imagery classification, in: *2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2023, pp. 3789–3794. doi:10.1109/SMC53992.2023.10394657.

- [15] X.-C. Zhong, Q. Wang, D. Liu, Z. Chen, J.-X. Liao, J. Sun, Y. Zhang, F.-L. Fan, EEG-DG: A multi-source domain generalization framework for motor imagery EEG classification, *IEEE Journal of Biomedical and Health Informatics* 29 (4) (2025) 2484–2495, first published online in 2024. doi:10.1109/JBHI.2024.3431230.
- [16] Y. Zheng, S. Wu, J. Chen, Q. Yao, S. Zheng, Cross-subject motor imagery electroencephalogram decoding with domain generalization, *Bioengineering* 12 (5) (2025) 495. doi:10.3390/bioengineering12050495.
- [17] P. Zanini, M. Congedo, C. Jutten, S. Said, Y. Berthoumieu, Transfer learning: A riemannian geometry framework with applications to brain–computer interfaces, *IEEE Transactions on Biomedical Engineering* 65 (5) (2018) 1107–1116. doi:10.1109/TBME.2017.2742541.
- [18] H. He, D. Wu, Transfer learning for brain–computer interfaces: A euclidean space data alignment approach, *IEEE Transactions on Biomedical Engineering* 67 (2) (2020) 399–410. doi:10.1109/TBME.2019.2913914.
- [19] M. Long, Y. Cao, J. Wang, M. I. Jordan, Learning transferable features with deep adaptation networks, in: *Proceedings of the 32nd International Conference on Machine Learning*, Vol. 37 of *Proceedings of Machine Learning Research*, 2015, pp. 97–105.
- [20] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, V. Lempitsky, Domain-adversarial training of neural networks, *Journal of Machine Learning Research* 17 (59) (2016) 1–35.
- [21] X. Tang, X. Zhang, Conditional adversarial domain adaptation neural network for motor imagery EEG decoding, *Entropy* 22 (1) (2020) 96. doi:10.3390/e22010096.
- [22] D. Wu, M. Tang, S. Liu, HADANet: Hybrid attentive domain adaptation for cross-subject motor imagery EEG decoding, *Cognitive Neurodynamics* 20 (1) (2026) 137. doi:10.1007/s11571-026-10510-7.
- [23] D. Zhang, H. Li, J. Xie, Unsupervised and semi-supervised domain adaptation networks considering both global knowledge and prototype-based local class information for motor imagery classification, *Neural Networks* 179 (2024) 106497. doi:10.1016/j.neunet.2024.106497.

- [24] F. M. Carlucci, L. Porzi, B. Caputo, E. Ricci, S. Rota Bulò, AutoDIAL: Automatic DomaIn alignment layers, in: Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2017, pp. 5067–5075.
- [25] J. Liang, D. Hu, J. Feng, Do we really need to access the source data? source hypothesis transfer for unsupervised domain adaptation, in: Proceedings of the 37th International Conference on Machine Learning, Vol. 119 of Proceedings of Machine Learning Research, 2020, pp. 6028–6039.
- [26] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, T. Darrell, Tent: Fully test-time adaptation by entropy minimization, in: International Conference on Learning Representations, 2021.
- [27] V. Jayaram, A. Barachant, MOABB: Trustworthy algorithm benchmarking for BCIs, *Journal of Neural Engineering* 15 (6) (2018) 066011. doi:10.1088/1741-2552/aadea0.
- [28] R. Leeb, F. Lee, C. Keinrath, R. Scherer, H. Bischof, G. Pfurtscheller, Brain–computer communication: Motivation, aim, and impact of exploring a virtual apartment, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 15 (4) (2007) 473–482. doi:10.1109/TNSRE.2007.906956.
- [29] M. Tangermann, K.-R. Müller, A. Aertsen, N. Birbaumer, C. Braun, C. Brunner, R. Leeb, C. Mehring, K. J. Miller, G. R. Müller-Putz, G. Nolte, G. Pfurtscheller, H. Preissl, G. Schalk, A. Schlögl, C. Vidaurre, S. Waldert, B. Blankertz, Review of the BCI competition IV, *Frontiers in Neuroscience* 6 (2012) 55. doi:10.3389/fnins.2012.00055.
- [30] B. Zhou, X. Wu, Z. Lv, L. Zhang, X. Guo, A fully automated trial selection method for optimization of motor imagery based brain–computer interface, *PLOS ONE* 11 (9) (2016) e0162657. doi:10.1371/journal.pone.0162657.
- [31] F. Lotte, Signal processing approaches to minimize or suppress calibration time in oscillatory activity-based brain–computer interfaces, *Proceedings of the IEEE* 103 (6) (2015) 871–890. doi:10.1109/JPROC.2015.2404941.